

NBA Player Impact Model

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1 Introduction

Throughout the illustrious history of the NBA, the salary of the player has been a hot topic of debate. Often, a player's production on the court does not match their pay. Whether it be a player outplaying or missing the mark on what they signed for, player salary can ignite the most heated arguments within a given fan base. So, the goal of this project is to pull statistics from different basketball data bases and try to predict what a player should be making based on their numbers. All of the statistics being used are from the 2025-26 NBA season. Disclaimer: Out of care for the players and the integrity of the game of basketball, it is important to note that statistics, no matter how advanced, are not always indicative of a player's contribution or value to a team. In a lot of cases there are players whose value goes beyond the stat sheet and their play on the court is the only way to see that.

2 Methodology

To create this model, advance statistics and regular per game statistics will be pulled from the "Basketball Reference" website [1]. The salary of each player will also be pulled from Basketball Reference [2]. The per game stats being used are points per game, assists per game, rebounds per game, games played, and blocks per game. The advanced stats being used are true shooting percentage, win shares, usage rate, and VORP. To create this prediction model, a `LinearRegression` object from the `sklearn.linear_model` Python package will be used. The data will be split using the standard 80-20 training-test split (i.e. about 80% of our data will be used to train our model and the other 20% will be used to test it).

2.1 Stats and Advanced Stats

A little bit more on the stats being used:

1. Points per game (PPG): Average points scored per game.
2. Assists per game (AST): Average assists dished per game.
3. Rebounds per game (REB): Average rebounds grabbed per game.
4. Blocks per game (Blocks): Average blocked shots per game.
5. Games played: Number of games played in the 2025 - 26 season (Out of 82).
6. True Shooting Percentage (TS%): Measures overall shooting efficiency including three point percentage, free throw percentage, and overall field goal percentage.
7. Win Shares (WS): The number of wins a player contributes by themselves.
8. Usage rate (USG%): An estimation of the percentage of plays that a player is directly involved in.
9. Value over Replacement Player (VORP): The extra points that a player provides over a minimum salary player bench player Per-100-possession [3]

2.2 Data Cleaning

Basketball Reference has players stats and salaries on different pages, so in order for this model to work, the names of the player had to be cleaned. To do this, I created a function which took in all player names and put them in lower case and had all hyphens removed. For example, "Shai Gilgeous-Alexander" became "shai gilgeousalexander." Player salaries were also cleaned by removing all commas and dollar signs then making the salary into a int instead of a string. All missing salaries and duplicates were dropped from the data set. Once this was done the players' stats were matched to their salaries via their cleaned names and I merged the salary and stats dataframes into one.

2.3 The Model

To create salary predictions, a linear regression was trained using the stats previously stated as the features and salary as the label. The model was trained on an 80-20 split meaning 80% of our players were used to train the model and the other 20% were used to evaluate it. In order to see if the regression was reliable I used Mean Absolute error (MAE), Root Mean Square error (RMSE), and R^2 .

3 Results

3.1 Feature Correlation

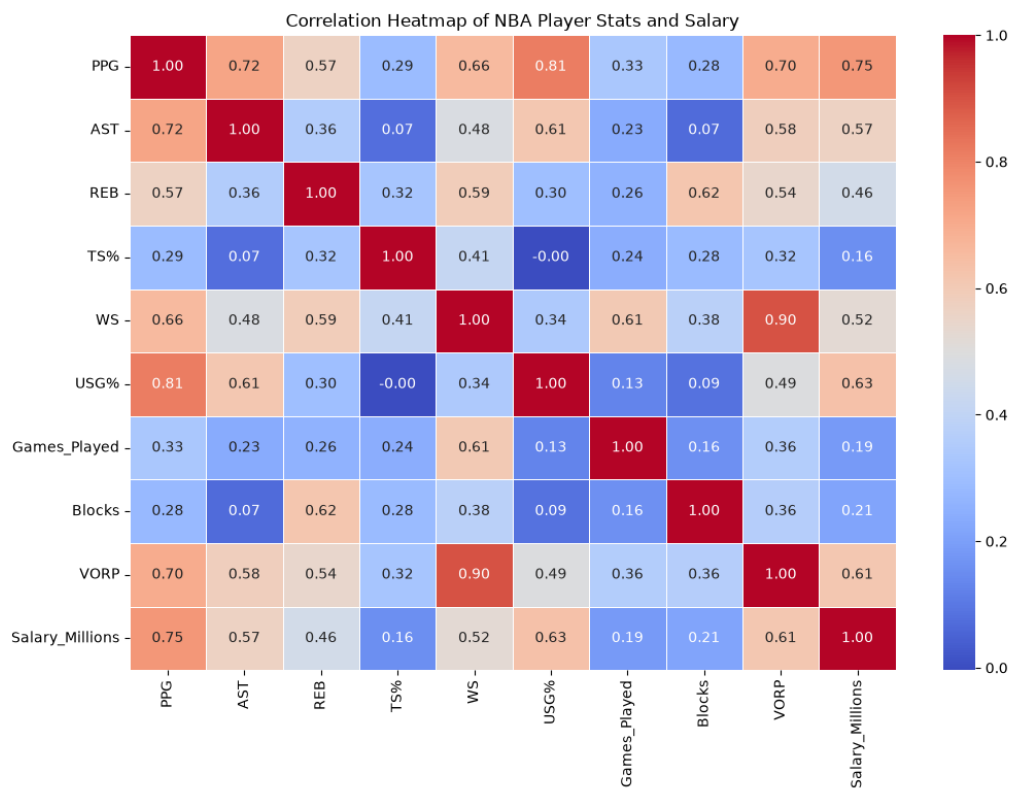


Figure 1: Correlation heat map of player statistics and salary. This shows that salary is most correlated with PPG, USG%, VORP, and AST.

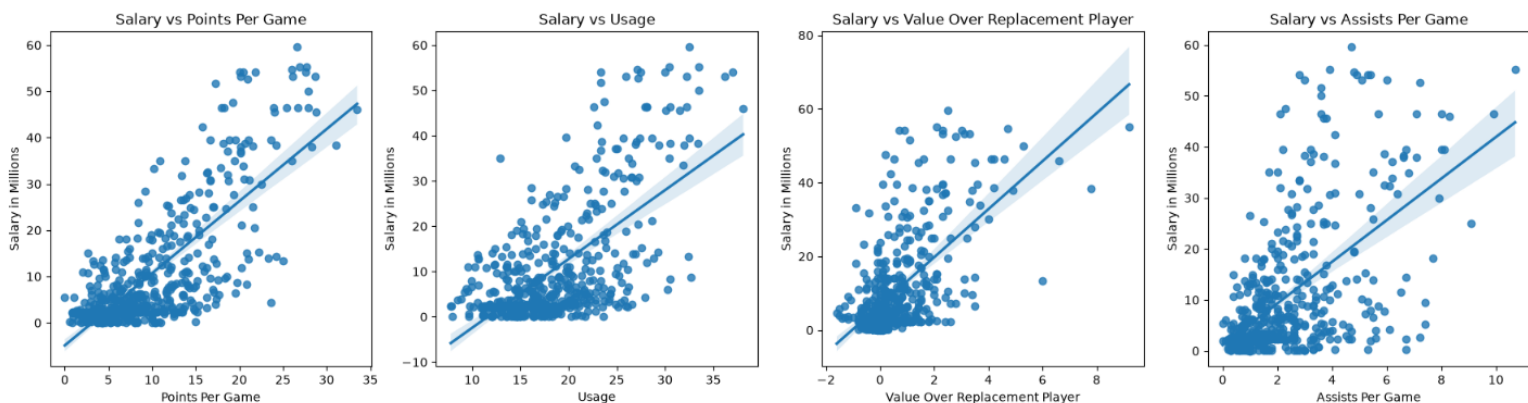


Figure 2: Here are the regression plots for salary compared to PPG, USG%, VORP, and AST.

3.2 Predicted Salary

After training the model, each of the 496 players' stats was plugged in and a predicted salary for each was produced. All 496 players predicted salaries are available in the NBBAIMPACT2.ipynb file. I will only highlight a few notable ones. First, the player with the largest salary difference was Spurs star Victor Wembanyama. His salary for the 2025-26 season was \$13,376,880 while the model predicted a salary of \$39,597,337. Next, the player with the smallest salary difference was 76er's forward Paul George. His salary for the 2025-26 season was \$51,666,090, while the model predicted a salary of \$22,477,428. Last, a few notable players will be listed below in the form of (Player, Actual salary, Predicted Salary, Salary Difference)

1. Stephen Curry, \$59,606,817, \$34,961,257, \$ - 24,645,560
2. LeBron James, \$52,627,153, \$29,069,243, \$ - 23,557,910
3. Kevin Durant, \$54,708,609, \$35,298,813, \$ - 19,409,796
4. Jordan Poole, \$31,848,215, \$12,840,289, \$ - 19,007,926
5. OG Anunoby, \$39,568,966, \$20,598,216, \$ - 18,970,750
6. James Harden, \$39,446,090, \$32,026,458, \$ - 7,419,632
7. Nikola Jokić, \$55,224,526, \$53,350,618, \$ - 1,873,908
8. Dillon Brooks, \$21,124,110, \$21,710,131, \$586,021
9. Nickeil Alexander-Walker, \$15,161,800, \$23,882,042, \$8,720,242
10. Austin Reaves, \$13,937,574, \$28,729,758, \$14,792,184

4 Discussion

4.1 Feature Correlation

In general, the features with the highest correlation with salary have possible explanations. First, If a player has a high PPG average, they will be paid well. The NBA is built on putting the ball in the basket, so if a player can do that well it will be highly sought after by an organization. This goes for AST as well. Players who can create shots which lead to baskets for their teammates are also contributing to putting the ball in the basket. As for usage rate, players who are the most elite (and highly paid) will typically have most plays called for them and will have the ball in their hands the majority of the play. Moving on to VORP, a player who performs much better than minimum contract player, i.e a high VORP, it makes sense for them to be paid a large sum of money. Players who have a much higher production than those on the bench will usually be those upper echelon tier of athletes who are paid well for their ability.

4.2 Predicted Salary

5 Conclusion

Overall, the predictions produced by the model seem to be accurate. For example, Victor Wembanyama has outplayed his contract by a long shot. This season Wembanyama won the defensive player of the year award, was second in MVP voting, and lead his team to the finals. Not only that but his numbers back it up as he averaged 25 PPG, 11.5 RPG, 3.1 AST, and 3.1 BPG. Another example is Nikeil Alexander-Walker. The reigning MIP had a huge jump in PPG going from 9.4 to 20.8. He was a leading force in the Atlanta Hawks push in the playoffs. On the other end of the spectrum, the most overpaid player predicted by the model was Paul George. This makes sense as George has seen a significant regression in his production since joining the Philadelphia 76ers. His PPG has dropped off from about 23 to 16 PPG since his last season with the Clippers.

In terms of error, the model had an $R^2 = 0.68$ and $RMSE = 8.69788$. While the $RMSE$ was not very high, the R^2 was a bit low. In the future, something that can help the model is creating separate regressions for each position. In basketball, each position is expected to have different types of productions. For example, a center or power forward is expected to grab rebounds and block shots while guards are expected to score and assist. So, with the current model, that difference may not be as clear and might cloud the regression. A player like Wembanyama may be favored as he dominates all statistical categories while a player like Stephen Curry may not average as many rebounds or blocks since he is a guard. This would force the model to predict that Curry is overpaid which is most definitely not the case.

References

- [1] Kubatko, Justin (2026). *2025-26 NBA Player Stats: Per Game*. Basketball-Reference.com
- [2] Kubatko, Justin (2026). *Contracts*. Basketball-Reference.com
- [3] Kubatko, Justin (2026). *Glossary* Basketball-Reference.com